

Abdul-Rahman Al-Zawawi

Detroit, MI

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Education

Wayne State University

Bachelor of Science in Biomedical Engineering

Detroit, MI

Expected Graduation: May 2027

Washtenaw Community College

Associate in General Math and Science

Ann Arbor, MI

Jan 2020 – May 2023

Projects

Artificial Hydraulic Heart Biomedical Engineering Design

Aug 2024 – May 2025

- Designed an artificial heart concept with a hydraulic actuation mechanism using AutoCAD for mechanical modeling
- Fabricated a 3D-printed prototype and integrated a motor-driven pumping system to simulate controlled fluid flow
- Modeled and evaluated **10+** CAD components, optimizing hydraulic flow paths and mechanical assembly
- Evaluated system performance through water flow testing and prepared detailed technical design documentation

Biomechanical Leg Analysis & Signal Processing

Jan 2025 – May 2025

- Constructed a custom 3D leg model in AutoCAD and analytically calculated its moment of inertia and dynamic properties
- Instrumented a physical leg with EMG sensors, collected swing-motion data, and imported data into MATLAB for review
- Acquired and processed **1,000+** EMG and motion data points using MATLAB for quantitative time-series signal analysis
- Validated analytical predictions by comparing theoretical dynamics with experimentally measured sensor-based results

3D Modeling Club

Jan 2025 – Apr 2025

- Co-founded a **20+** member student team focused on learning AutoCAD basics through hands-on design workshops
- Led competitive team-based “battling tops” design challenge, managing initial fabrication to final tournament execution
- Mentored team members through AutoCAD iterations, troubleshooting design issues, and improving technical skills
- Coordinated design reviews, fabrication timelines, and project execution to ensure successful event completion

Remote-Controlled Vehicle

Aug 2023 – Jan 2024

- Designed a remote-controlled car, creating full 3D CAD model and fabricating components with 3D printing for prototyping
- Configured Arduino, motors, and electronics, ensuring mechanical fit, wiring, and system functionality across subsystems
- Modeled and assembled **15+** mechanical and electronic components, integrating motors, Arduino, and chassis
- Developed a C++ control program and a phone-based remote interface to control vehicle motion wirelessly and reliably

Arduino Digital Clock

Nov 2022 – Jan 2023

- Built an Arduino digital clock using a DS1302 RTC module and I2C LCD display for real-time date output and timekeeping
- Programmed C++ firmware to interface with the Real-Time Clock and format time display with custom AM/PM logic
- Tested **8+** hardware connections (RTC, LCD, power), debugging signal integrity, stability, and I2C communication errors
- Verified battery operation and perpetual timekeeping over **100+** hours, conducting repeated functional and reliability tests

Professional Experience

Warrior Consulting Group

Detroit, MI

Consulting team lead for Nudena

Sep 2025 – Dec 2025

- Analyzed RFQ-to-production manufacturing systems to identify inefficiencies and scalability constraints across suppliers
- Evaluated **20+** manufacturers, creating RFQ, BOM, legal documents and insight reports on material and tooling decisions

The Home Depot

Ypsilanti, MI

Order Fulfillment / Sales Associate

Jun 2023 – Aug 2024

- Managed order fulfillment and inventory operations in a high-volume retail environment with time sensitive demands
- Operated forklifts and communicated technical product information to support efficient store operation and customer need

Starbucks

Belleville, MI

Team Lead / Barista

Dec 2020 – Nov 2021

- Managed team operation in fast-paced conditions, ensuring workflow efficiency, quality, and safety compliance
- Trained and supervised employees while handling high-volume order processing with accuracy and consistency

Skills & Interests

CAD & Design: AutoCAD, Siemens NX, Solidworks, tolerance-aware modeling

Programming & Analysis: C/C++, MATLAB, Python, signal & data analysis

Hardware & Prototyping: Arduino, motor integration, Oscilloscope, sensors, 3D printing

Languages: Fluent in Arabic

Interests: Biomechanics, BJJ, Ice Skating, MMA, Powerlifting, Robotics, Snowboarding, Technical Analysis, 3D Modeling